Minimum Viable Scale and Local Activation in Nonconcave Growth

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Abstract

We study a Ramsey economy with endogenous labor supply and a CES technology homogeneous of degree θ. In the fold regime $\theta > 1$ and $\sigma < 1$, the steady-state Euler restriction $F_K = D$ and homogeneity reduce the interior steady-state problem to zeros of a scalar diagnostic $G(r; D)$ in the capital–labor ratio. The reduced marginal product of capital is hump-shaped, which yields a closed-form minimum viable labor scale

$$L_{\min} = \left(\frac{D}{\phi_K^{\max}} \right)^{1/(\theta-1)}$$

and generically permits multiple interior steady states above that scale. The paper's main global result is therefore static. On the dynamic side we proceed locally. After solving labor out of the planner's problem, we formulate reduced Bellman equations on compact admissible branch neighborhoods, obtain continuous branch values, and study the geometry of the overlap through activation sets, first activation pockets, and the leftmost interior crossing $K^{(1)}(D)$ of the branch-value gap. When the activation set is boundary-attached, the minimal threshold $K_S^{\min}(D)$ is retained only as a branch-boundary diagnostic, while the economic interpretation falls on $K^{(1)}(D)$ and the width of the first activation pocket. In the benchmark, the two steady states are far apart in stationary terms, yet the leftmost interior crossing lies only slightly above the lower edge of the high branch. We also formalize the distinction between the fold threshold D_F and the final saddle-node threshold D_{SN}, so recovery is a two-condition problem involving both wedge restoration and entry into the relevant local activation region. An unrestricted reduced-Bellman solve on a wide interval is used as validation rather than as a global theorem: it shows a branch envelope close to the unrestricted planner value, tiny capital-policy disagreement, larger labor-policy disagreement, and a thin negative envelope sliver. Measured against the paper's actual local claim, the unrestricted leftmost crossing differs from the branch crossing by only about one-tenth of a first-pocket width, while the maximum envelope gap is below one-tenth of one percent in permanent consumption-equivalent terms. Appendix B makes the corresponding decision-quality diagnostics and rerun standards explicit. The quantitative message is therefore large stationary separation, a narrow local first-crossing region, and disciplined rather than global claims about hysteresis in nonconcave Ramsey–CES dynamics.

1 Introduction

The canonical one-sector growth model relies on constant or decreasing returns to scale because concavity keeps both the steady-state and dynamic sides tractable. Once returns become increasing, that structure breaks. Global concavity disappears, the steady-state problem can admit multiple interior solutions, and threshold dynamics become possible. Those possibilities are not merely technical curiosities. They sit naturally in environments with production complementarities, indivisibilities, and scale effects of the kind emphasized in development, trade, and macroeconomics [Murphy et al., 1989, Azariadis and Drazen, 1990, Krugman, 1991, Basu and Fernald, 1997, Antweiler and Trefler, 2002]. At the same time, empirical surveys continue to find wide scope for capital-labor complementarity and substantial heterogeneity in substitution elasticities [Knoblach and Stockl, 2020, Gechert et al., 2022]. The obstacle is therefore not the economic relevance of increasing returns, but the lack of tractable theory once nonconcavity is admitted.

This paper shows that a Ramsey economy with endogenous labor supply and a CES technology homogeneous of degree θ remains surprisingly tractable in the relevant nonconcave region. The key observation is that steady state combines the Euler restriction $F_K = D$, where $D \equiv \beta^{-1} - (1-\delta)$ is the user-cost wedge, with homogeneity. That combination lets us write the steady-state problem in terms of the capital-labor ratio $r \equiv K/L$. In the fold regime, defined by $\theta > 1$ and $\sigma < 1$, the reduced marginal product of capital is hump-shaped in r. The hump creates a fold in the iso-user-cost locus, generates a minimum viable labor scale in closed form, and makes multiplicity a one-dimensional root-finding problem.

The paper makes four contributions.

1. **A scalar steady-state diagnostic.** Interior steady states correspond to zeros of a scalar function $G(r; D)$ along the iso-user-cost locus $F_K = D$. This collapses an otherwise nonconcave three-equation steady-state system to a one-dimensional problem.

2. **A closed-form minimum viable scale.** In the fold regime, the Euler restriction implies

$$L_{\min} = \left(\frac{D}{\phi_K^{\max}} \right)^{1/(\theta-1)},$$

   below which no interior steady state can exist. The exponent $1/(\theta-1)$ is a policy amplifier: when increasing returns are mild, small changes in the wedge translate into large changes in minimum viable scale.

3. **A disciplined local dynamic theory.** After solving labor out period by period, we define reduced Bellman equations on compact admissible local neighborhoods around the coexisting steady states. A uniform strict-feasibility theorem gives a continuous reduced return, branch-restricted Bellman problems are well posed, activation regions admit a pocket decomposition, and branch values vary locally Lipschitz-continuously with the wedge. The paper's central dynamic theorem is a local persistence result for the *leftmost interior crossing* $K^{(1)}(D)$ under a

single-crossing condition. When the activation set is boundary-attached, $K_S^{\min}(D)$ is reported only as a branch-boundary diagnostic rather than as an intrinsic economic threshold.

4. **A two-threshold wedge taxonomy with explicit validation objects.** The steady-state wedge analysis distinguishes the fold threshold D_F, at which opposite-branch multiplicity disappears, from the final saddle-node threshold D_{SN}, at which the last interior steady states vanish. Recovery is therefore two-dimensional: it requires both $D < D_{SN}$ and an initial capital stock inside the relevant local activation region. On the numerical side, the paper organizes validation around branch Bellman residuals, an unrestricted reduced-Bellman envelope gap, policy discrepancies, and the decision-quality objects defined in Appendix B rather than around Euler equations alone.

Two scope remarks are important. First, the paper's *main theorem-level contribution is static*. The dynamic side should be read as a disciplined local activation theory plus validation objects, not as a global extension of Dechert–Nishimura. Second, the benchmark is illustrative rather than structurally estimated. Total factor productivity is chosen so that minimum viable scale is economically relevant, and the quantitative section should be read as a worked example rather than as full structural identification.

The paper sits next to, but does not reproduce, the classic nonconcave-growth literature. Skiba [1978] gives the original threshold intuition in a continuous-time convex-concave growth model. Majumdar and Mitra [1982] and Dechert and Nishimura [1983] establish stronger discrete-time results for canonical one-sector nonconcave models, including critical-capital thresholds and monotone optimal paths. Askenazy and Le Van [1999] study poverty-trap dynamics in a convex-concave growth model with external borrowing. Akao et al. [2011] show that the critical capital stock in the Dechert-Nishimura environment is continuous and monotone in the discount factor, and Akao et al. [2025] obtain the analogous continuous-time result for a convex-concave technology. More generally, Kamihigashi and Roy [2007] show how difficult optimal growth becomes once convexity and smoothness are lost. Relative to that literature, our dynamic contribution is intentionally narrower and more transparent: it is a local activation-and-hysteresis theory for a Ramsey-CES environment with endogenous labor and increasing returns, not a global extension of Dechert-Nishimura.

The computational side is also aligned with the nonconcave literature. Fella [2014] and Druedahl and Jorgensen [2017] emphasize that threshold-sensitive nonconcave problems require methods that can handle kinks, local nonmonotonicities, and irregular grids. Judd [1992] and Santos and Vigo-Aguiar [1998] show why numerical accuracy should be assessed through Bellman and policy errors, not only through iteration convergence. Those lessons are central here because threshold accuracy, not merely Bellman convergence, is the object of interest.

The rest of the paper is organized as follows. Section 2 presents the model. Section 3 develops the scalar steady-state diagnostic. Section 4 derives the fold regime, minimum viable scale, and multiplicity conditions. Section 5 provides a benchmark quantitative illustration. Section 6 develops the reduced dynamic problem, local activation geometry, and two-threshold recovery logic. Section

7 concludes. Appendix A collects the proofs for the dynamic propositions. Appendix B documents the numerical implementation and verification diagnostics.

Output is produced by a CES technology homogeneous of degree θ:

$$F(K,L) = A\left[\omega K^{\xi} + (1-\omega)L^{\xi}\right]^{\theta/\xi}, \qquad A>0,\ \omega\in(0,1),\ \xi\in\mathbb{R}\setminus\{0\},\ \theta>0. \tag{2.3}$$

Let

$$Z(K,L) \equiv \omega K^{\xi} + (1-\omega)L^{\xi}. \tag{2.4}$$

The marginal products are

$$F_K(K,L) = A\theta\omega Z(K,L)^{\theta/\xi-1}K^{\xi-1}, \tag{2.5}$$

$$F_L(K,L) = A\theta(1-\omega)Z(K,L)^{\theta/\xi-1}L^{\xi-1}. \tag{2.6}$$

The elasticity of substitution is $\sigma = 1/(1-\xi)$, so $\sigma<1$ is equivalent to $\xi<0$.

For interior allocations, the planner's first-order conditions are

$$\frac{1-\eta}{\eta}\frac{C_t}{1-L_t} = F_L(K_t,L_t), \tag{2.7}$$

$$\frac{1}{C_t} = \beta\frac{1}{C_{t+1}}\left[F_K(K_{t+1},L_{t+1}) + 1 - \delta\right]. \tag{2.8}$$

Define the user-cost wedge

$$D \equiv \beta^{-1} - (1-\delta) > 0. \tag{2.9}$$

In any interior stationary allocation, (2.8) implies

$$F_K(K,L) = D. \tag{2.10}$$

This stationary Euler restriction is the starting point for the paper's steady-state reduction.

3 Scalar Steady-State Reduction

Define the capital-labor ratio $r \equiv K/L$ and the reduced CES index

$$Z(r) \equiv \omega r^{\xi} + (1-\omega). \tag{3.1}$$

Homogeneity of degree θ gives the decomposition

$$F_K = L^{\theta-1}\phi_K(r), \qquad F_L = L^{\theta-1}\phi_L(r), \qquad F = L^{\theta}y(r), \tag{3.2}$$

where

$$\phi_K(r) \equiv A\theta\omega Z(r)^{\theta/\xi-1}r^{\xi-1}, \tag{3.3}$$

$$\phi_L(r) \equiv A\theta(1-\omega)Z(r)^{\theta/\xi-1}, \tag{3.4}$$

$$y(r) \equiv AZ(r)^{\theta/\xi}. \tag{3.5}$$

The decomposition separates scale, which is carried by powers of L, from composition, which is entirely summarized by r.

We focus on the increasing-returns case $\theta > 1$. Then the stationary Euler condition (2.10) implies

$$L(r) = \left(\frac{D}{\phi_K(r)}\right)^{1/(\theta-1)}, \qquad K(r) = rL(r), \qquad C(r) = L(r)^\theta y(r) - \delta r L(r). \tag{3.6}$$

Define the scalar steady-state diagnostic

$$G(r) \equiv \frac{1-\eta}{\eta}\frac{C(r)}{1-L(r)} - L(r)^{\theta-1}\phi_L(r). \tag{3.7}$$

Proposition 3.1 (Scalar steady-state diagnostic)**.** *Suppose $\theta > 1$. A triple (C, K, L) with $C > 0$, $K > 0$, and $L \in (0,1)$ is an interior steady state if and only if there exists $r > 0$ such that $L(r) \in (0,1)$, $C(r) > 0$, and $G(r) = 0$.*

Proof. If (C, K, L) is an interior steady state, define $r = K/L$. Then (2.10) implies $L^{\theta-1}\phi_K(r) = D$, hence $L = L(r)$ and $K = K(r)$. The resource constraint gives $C = C(r)$, and the intratemporal condition (2.7) reduces to $G(r) = 0$. Conversely, if $r > 0$ satisfies $L(r) \in (0,1)$, $C(r) > 0$, and $G(r) = 0$, then setting $(K, L, C) = (K(r), L(r), C(r))$ satisfies (2.2), (2.7), and (2.10) by construction. □

Proposition 3.1 is the paper's main steady-state simplification. Interior steady states are no longer found by solving a three-dimensional nonlinear system. They are found by locating zeros of a scalar diagnostic along the iso-user-cost locus $F_K = D$.

4 The Fold Regime, Minimum Viable Scale, and Multiplicity

Definition 4.1 (Fold regime)**.** The fold regime is defined by $\theta > 1$ and $\xi < 0$, equivalently $\sigma < 1$.

The fold regime combines increasing returns with capital-labor complementarity. That combination makes the reduced marginal product of capital nonmonotone in the capital-labor ratio.

4.1 Hump shape of the reduced marginal product

Lemma 4.2 (Boundary behavior)**.** *In the fold regime, $\phi_K(r) \to 0$ as $r \downarrow 0$ and as $r \uparrow \infty$.*

Proof. As $r \downarrow 0$, we have $r^\xi \to \infty$, so $Z(r) \sim \omega r^\xi$ and therefore $\phi_K(r) \sim A\theta\omega^{\theta/\xi} r^{\theta-1} \to 0$ because $\theta - 1 > 0$. As $r \uparrow \infty$, we have $r^\xi \to 0$, so $Z(r) \to 1-\omega$ and $\phi_K(r) \sim A\theta\omega(1-\omega)^{\theta/\xi-1} r^{\xi-1} \to 0$ because $\xi - 1 < 0$. □

Lemma 4.3 (Unique interior maximum)**.** *In the fold regime, ϕ_K has a unique maximizer $r^* > 0$ satisfying*

$$(r^*)^\xi = \frac{(1-\xi)(1-\omega)}{\omega(\theta-1)}. \tag{4.1}$$

Let $\phi_K^{\max} \equiv \phi_K(r^*) > 0$.

Proof. Differentiating $\log \phi_K(r)$ gives

$$\frac{d}{dr} \log \phi_K(r) = \left(\frac{\theta}{\xi} - 1\right) \frac{\omega \xi r^{\xi-1}}{Z(r)} + \frac{\xi - 1}{r}.$$

Setting this derivative equal to zero yields

$$(\theta - \xi)\omega r^{\xi} = (1 - \xi)(\omega r^{\xi} + 1 - \omega),$$

which simplifies to (4.1). Since $r \mapsto r^{\xi}$ is strictly monotone for $\xi \neq 0$, the critical point is unique. Lemma 4.2 implies that it is the global maximum. □

4.2 Minimum viable scale

Proposition 4.4 (Minimum viable scale)**.** *In the fold regime, any interior steady state must satisfy*

$$L \geq L_{\min} \equiv \left(\frac{D}{\phi_K^{\max}}\right)^{1/(\theta-1)}. \tag{4.2}$$

If $L_{\min} \geq 1$*, no interior steady state exists.*

Proof. At any interior steady state, (2.10) and (3.2) give $D = L^{\theta-1}\phi_K(r) \leq L^{\theta-1}\phi_K^{\max}$. Rearranging yields (4.2). If $L_{\min} \geq 1$, then no admissible labor choice $L \in (0, 1)$ can satisfy the inequality. □

Corollary 4.5 (Wedge amplifier)**.** *The elasticity of minimum viable scale with respect to the wedge is*

$$\frac{\partial \log L_{\min}}{\partial \log D} = \frac{1}{\theta - 1}. \tag{4.3}$$

Hence mild increasing returns are the most dangerous case: as $\theta \downarrow 1$*, small wedge changes produce large movements in* $L_{\min}$*.*

4.3 Fold geometry and multiplicity

Proposition 4.6 (Two branches)**.** *If* $L_{\min} < 1$*, then for each* $L \in (L_{\min}, 1)$ *the equation*

$$\phi_K(r) = \frac{D}{L^{\theta-1}} \tag{4.4}$$

has exactly two solutions, $r^-(L) < r^* < r^+(L)$*. At* $L = L_{\min}$ *the unique solution is* $r = r^*$*. Near the fold,*

$$r^{\pm}(L) \approx r^* \pm c\sqrt{L - L_{\min}}, \qquad c = \sqrt{\frac{2\phi_K^{\max}(\theta - 1)}{|\phi_K''(r^*)| L_{\min}}}. \tag{4.5}$$

Proof. The hump shape from Lemmas 4.2 and 4.3 implies that the horizontal line $D/L^{\theta-1}$ intersects ϕ_K twice whenever it lies below $\phi_K^{\max}$, once on each side of r^*. At $L = L_{\min}$ the line is tangent to the hump. A second-order Taylor expansion of ϕ_K around r^* then yields the square-root approximation. □

Proposition 4.7 (Steady-state multiplicity)**.** *Suppose $L_{\min} < 1$, consumption is positive on the feasible locus, and*

$$G(r^*) < 0. \tag{4.6}$$

Then G has at least one root on each branch of the fold, yielding a low-r steady state and a high-r steady state.

Proof. On each branch, as r approaches the boundary where $L(r) \uparrow 1$, the term $C(r)/(1 - L(r))$ diverges and therefore $G(r) \to +\infty$. Condition (4.6) gives $G(r^*) < 0$. The intermediate value theorem then implies at least one zero on each branch. □

Remark 4.8. Condition (4.6) is sufficient but not necessary for multiplicity. Near the fold threshold it is possible for a same-branch pair of steady states to survive briefly after opposite-branch multiplicity disappears. That distinction becomes important in the wedge taxonomy of Section 6.

5 Benchmark Quantitative Illustration

This section provides a worked quantitative example. Its role is not to estimate the model structurally but to show that the steady-state geometry and the local dynamic objects are quantitatively meaningful in a plausible neighborhood of standard annual parameter values.

5.1 Benchmark calibration and CRS comparison

Table 1 reports the benchmark. The returns-to-scale parameter is set to $\theta = 1.25$, the elasticity of substitution to $\sigma = 2/3$ (equivalently $\xi = -0.5$), the discount factor to $\beta = 0.96$, depreciation to $\delta = 0.08$, the consumption weight to $\eta = 0.40$, and the distribution parameter to $\omega = 0.35$. Total factor productivity $A = 0.057$ is chosen so that minimum viable scale is economically relevant rather than trivial.

Table 1: Benchmark calibration

Parameter	Value	Interpretation
Returns to scale θ	1.25	Mild but nontrivial increasing returns
Elasticity of substitution σ (ξ)	2/3 (-0.5)	Conservative complementarity benchmark
Discount factor β	0.96	Standard annual value
Depreciation δ	0.08	Standard annual value
Consumption weight η	0.40	Consistent with labor near one-third
Capital distribution ω	0.35	Rough capital-share discipline
TFP A	0.057	Chosen so that $L_{\min}$ is relevant

It is useful to compare the benchmark with the constant-returns case. Setting $\theta = 1$ and leaving all other parameters unchanged recovers the standard Ramsey-CES economy. Then the scale term $L^{\theta-1}$ disappears from the Euler restriction, the fold vanishes, and the model has a unique steady state:

$$K^* = 0.0270, \qquad L^* = 0.2932, \qquad Y^* = 5.14 \times 10^{-3}, \qquad C^* = 2.98 \times 10^{-3}.$$

The fold geometry, minimum viable scale, and multiplicity are therefore pure consequences of increasing returns interacting with complementarity.

5.2 Benchmark steady states and the amplifier

The benchmark increasing-returns economy has two interior steady states. Table 2 reports the ratio, capital, labor, output, consumption, and marginal product of capital at each steady state.

Table 2: Interior steady states in the benchmark increasing-returns calibration

	r	K	L	Y	C	F_K
Low-scale steady state	0.00513	0.00135	0.26269	1.49×10^{-4}	4.08×10^{-5}	0.12167
High-scale steady state	0.02448	0.00834	0.34066	1.05×10^{-3}	3.80×10^{-4}	0.12167

The upper steady state is much larger in economic terms: $K_H/K_L \approx 6.19$, $Y_H/Y_L \approx 7.05$, and $L_H/L_L \approx 1.30$. The stationary reallocation between the two regimes is therefore substantial.

The wedge amplifier is also economically meaningful. At the benchmark $\theta = 1.25$, the elasticity in (4.3) equals four. A 10% increase in the wedge raises $L_{\min}$ by roughly 46%, a 25% increase nearly doubles it, and a 40% increase pushes the minimum viable scale close to one.

Holding $L_{\min} = 0.25$ fixed and recalibrating A cell by cell, opposite-branch multiplicity first appears for relatively mild increasing returns across a broad range of complementarity assumptions: the threshold value of θ falls from about 1.27 at $\sigma = 0.40$ to about 1.18 at $\sigma = 0.90$. Thus the fold regime is not a knife-edge feature of extreme parameter choices.

5.3 Neighborhood sensitivity

Table 3 reports one-at-a-time perturbations around the benchmark. The point of the exercise is modest: to show that the model's main comparative-static message is robust in a neighborhood of the benchmark. Across the cases shown, the economy continues to display two interior steady states and a strict wedge interval $D_F < D_{SN}$. The size of the stationary gap moves materially, but the basic viability and hysteresis logic survives.

The benchmark therefore enters the dynamic section with a clear static message: the two steady states are far apart, the viability boundary is quantitatively relevant, and the wedge interval between fold loss and final disappearance is narrow but nontrivial. Section 6 asks whether the corresponding dynamic selection margin is also large or instead concentrated in a narrow local activation region.

Case	A	η	ω	$L_{\min}$	K_H/K_L	D_F/D_0	D_{SN}/D_0	$\Delta K_S^{\min}/K_L$
Lower A	0.056	0.40	0.350	0.271	3.94	1.011	1.017	–
Benchmark	0.057	0.40	0.350	0.252	6.19	1.023	1.031	0.95%
Higher A	0.058	0.40	0.350	0.235	8.84	1.036	1.044	–
Lower η	0.057	0.39	0.350	0.252	5.22	1.018	1.025	–
Higher η	0.057	0.41	0.350	0.252	7.21	1.029	1.036	–
Lower ω	0.057	0.40	0.345	0.228	10.16	1.042	1.050	–
Higher ω	0.057	0.40	0.355	0.278	3.17	1.006	1.012	–

Table 3: One-at-a-time sensitivity around the benchmark. The last column reports the minimal local activation threshold only for the benchmark branch construction. The neighboring rows remain purely static because the local threshold depends on the branch neighborhoods used in Section 6.

5.4 Empirical discipline and limits of the benchmark

The benchmark is designed to make the model's viability boundary quantitatively visible. It is not a structural estimate of the production technology. In particular, the chosen value of A should be interpreted as a reduced-form shifter of ecosystem productivity, coordination efficiency, or spillover intensity rather than as a normalization with empirical uniqueness. The right question for the quantitative section is therefore not whether the benchmark identifies technology pointwise, but whether the model's core objects — $L_{\min}$, the scalar diagnostic $G(r; D)$, the leftmost interior crossing $K^{(1)}$, and the wedge thresholds D_F and D_{SN} — remain informative in a neighborhood of empirically plausible parameter values. That is the standard the benchmark and its neighborhood sensitivity exercise are designed to meet.

5.5 Primitive positivity on the Euler locus

The multiplicity result in Proposition 4.7 invokes positivity of consumption on the feasible Euler locus. In the present CES environment that condition can be reduced to primitives rather than left as an auxiliary restriction.

Proposition 5.1 (Primitive consumption-positivity criterion on the Euler locus)**.** *Along the stationary Euler locus $L(r)^{\theta-1}\phi_K(r) = D$,*

$$C(r) = L(r)\, r \left[\frac{D}{\theta\omega}(\omega + (1-\omega)r^{-\xi}) - \delta \right]. \tag{5.1}$$

Hence:

(i) *if $D \geq \delta\theta$, then $C(r) > 0$ for every $r > 0$;*

(ii) *if $D < \delta\theta$, then $C(r) > 0$ if and only if*

$$r > \bar{r}_C(D) \equiv \left[\frac{\omega}{1-\omega} \left(\frac{\delta\theta}{D} - 1 \right) \right]^{1/(-\xi)}. \tag{5.2}$$

Proof. Using $C(r) = L(r)^\theta y(r) - \delta r L(r)$ together with

$$y(r) = AZ(r)^{\theta/\xi}, \qquad \phi_K(r) = A\theta\omega Z(r)^{\theta/\xi - 1} r^{\xi - 1},$$

one obtains

$$y(r) = \frac{r\phi_K(r)}{\theta\omega}(\omega + (1-\omega)r^{-\xi}).$$

Substituting the Euler-locus identity $L(r)^{\theta-1}\phi_K(r) = D$ yields (5.1). Since $L(r)r > 0$, the sign of $C(r)$ is the sign of the bracketed term. If $D \geq \delta\theta$, then $\omega + (1-\omega)r^{-\xi} \geq \omega$ implies

$$\frac{D}{\theta\omega}(\omega + (1-\omega)r^{-\xi}) - \delta \geq \frac{D}{\theta} - \delta \geq 0,$$

and the inequality is strict for every $r > 0$ because $(1-\omega)r^{-\xi} > 0$. If $D < \delta\theta$, rearranging the strict inequality in (5.1) gives

$$r^{-\xi} > \frac{\omega}{1-\omega}\left(\frac{\delta\theta}{D} - 1\right),$$

which is equivalent to (5.2) because $-\xi > 0$. □

For the benchmark calibration, $D_0 = \beta^{-1} - (1-\delta) \approx 0.12167$ while $\delta\theta = 0.100$. Thus the positivity condition used in Proposition 4.7 is automatic on the entire feasible Euler locus in the benchmark. More generally, any calibration with $D \geq \delta\theta$ inherits the same simplification.

6 Reduced One-State Dynamics, Activation Correspondences, and Two-Threshold Hysteresis

Sections 3 and 4 deliver the paper's global result: the steady-state problem collapses to zeros of $G(r; D)$. The remaining question is dynamic and narrower: when two interior steady states coexist, which continuation is locally preferred on admissible neighborhoods, and what dynamic object can be reported without pretending to solve the unrestricted nonconcave planner problem globally? This section answers that question deliberately locally. After solving labor out of the planner's problem, we formulate a reduced one-state Bellman problem on compact admissible branch neighborhoods, compare the resulting branch values on their overlap, and organize the dynamic discussion around activation sets, first activation pockets, and especially the leftmost interior crossing $K^{(1)}(D)$ of the value gap. When the activation set is boundary-attached, $K_S^{\min}(D)$ remains useful as a branch-boundary diagnostic, but the economically interpretable object is $K^{(1)}(D)$ or, under Proposition 6.9, the upper endpoint of the first activation pocket. The section then links those local objects to the static wedge thresholds D_F and D_{SN} and validates them against an unrestricted reduced-Bellman solve on a wider interval. What follows is therefore a disciplined local dynamic theory, not a global Skiba theorem.

This section sits next to, but does not replicate, the global threshold results in Skiba (1978), Dechert and Nishimura (1983), and Akao, Kamihigashi, and Nishimura (2011, 2025). Relative to that literature, the contribution here is narrower and more transparent: a local activation theory for a Ramsey–CES environment with endogenous labor and increasing returns. On the numerical side, Fella (2014), Druedahl and Jorgensen (2017), Judd (1992), and Santos and Vigo-Aguiar (1998) explain why threshold-sensitive nonconcave problems should be validated through value and policy diagnostics rather than iteration convergence alone. That is the validation philosophy adopted below.

Throughout this section the user-cost wedge D is the primitive parameter. When a financing interpretation is useful, one may index the problem by a spread $s \geq 0$ through

$$\beta_s = \frac{\beta}{1+s}, \qquad D(s) = \beta_s^{-1} - (1-\delta),$$

but all theorems below are stated directly in terms of D.

6.1 Reduced one-state problem

Let

$$\beta(D) \equiv \frac{1}{D+1-\delta}. \tag{6.1}$$

For any current capital $K \geq 0$, labor choice $L \in [0,1)$, and next-period capital $K' \geq 0$, define

$$C(K,L,K') \equiv (1-\delta)K + F(K,L) - K'. \tag{6.2}$$

Let $B \equiv A(1-\omega)^{\theta/\xi}$. Since $\xi < 0$, output is globally bounded above by B, so the interval $[0, B/\delta]$ is absorbing from above. Fix compact intervals $I, I' \subset (0, B/\delta]$.

Definition 6.1 (Uniform strict feasibility)**.** *The pair (I, I') is uniformly strictly feasible if there exists $\bar{L} \in (0,1)$ such that*

$$c(I, I'; \bar{L}) \equiv \inf_{K \in I,\, K' \in I'} C(K, \bar{L}, K') > 0. \tag{6.3}$$

Uniform strict feasibility is stronger than pointwise feasibility, but it is the right local condition for a theory section: it produces a common finite benchmark payoff and forces all optimizing labor choices into a compact interior set. That, in turn, makes the reduced return clean.

Define the reduced return by

$$R(K,K') \equiv \sup_{L \in [0,1)} \{\eta \log C(K,L,K') + (1-\eta)\log(1-L)\}, \tag{6.4}$$

with the convention that the objective equals $-\infty$ whenever $C(K,L,K') \leq 0$.

Proposition 6.2 (Reduced return under uniform strict feasibility)**.** *Fix compact intervals $I, I' \subset (0, B/\delta]$ and suppose (I, I') is uniformly strictly feasible. Then there exists $\varepsilon \in (0,1)$ such that, for every $(K,K') \in I \times I'$, every maximizer in* (6.4) *lies in*

$$\Gamma_\varepsilon(K,K') \equiv \{L \in [0, 1-\varepsilon] : C(K,L,K') \geq \varepsilon\}. \tag{6.5}$$

Consequently: (i) *$R(K,K')$ is finite and continuous on $I \times I'$;* (ii) *the argmax correspondence in* (6.4) *is nonempty, compact-valued, and upper hemi-continuous; and* (iii) *for any continuous continuation value W, the reduced Bellman objective $(K,K') \mapsto R(K,K') + \beta(D)W(K')$ is continuous on $I \times I'$.*

Proposition 6.2 is where the local dynamic section becomes genuinely theorem-level. The logarithmic objective is not maximized on a globally compact labor set, so one needs more than a quick appeal to Berge. Uniform strict feasibility supplies the missing compactness by ruling out maximizing sequences that drive consumption to zero or labor to one.

6.2 Admissible local branch neighborhoods and branch values

The theory is developed on fixed branch neighborhoods over a compact wedge set.

Definition 6.3 (Admissible local branch neighborhoods)**.** *Let $\mathcal{D} \subset (0, \infty)$ be compact. A compact interval $I \subset (0, B/\delta]$ is admissible on $\mathcal{D}$ if* (i) *(I, I) is uniformly strictly feasible, and* (ii) *for every $D \in \mathcal{D}$ and every $K \in I$, there exists at least one $K' \in I$ with finite reduced return $R(K, K')$. A pair $(I^{\text{low}}, I^{\text{high}})$ is an admissible local branch pair on $\mathcal{D}$ if both intervals are admissible on $\mathcal{D}$ and their overlap*

$$J \equiv I^{\text{low}} \cap I^{\text{high}}$$

is nonempty.

In applications the intervals are chosen around the low and high steady states at the benchmark wedge so that the induced continuation preserves the intended local low-branch and high-branch interpretation. The definition above is deliberately weaker than a full basin theorem: it gives exactly what the Bellman arguments require without pretending to characterize global basins of attraction.

Given an admissible branch pair, define the branch-restricted Bellman operators

$$(T_D^j W)(K) \equiv \max_{K' \in I^j} \{R(K, K') + \beta(D) W(K')\}, \qquad j \in \{\text{low}, \text{high}\}. \tag{6.6}$$

Let V_D^j denote the unique fixed point of T_D^j on $C(I^j)$.

Proposition 6.4 (Branch-restricted Bellman problems)**.** *Fix a compact wedge set $\mathcal{D} \subset (0, \infty)$ and an admissible local branch pair $(I^{\text{low}}, I^{\text{high}})$ on $\mathcal{D}$. Then for each $D \in \mathcal{D}$ and each $j \in \{\text{low}, \text{high}\}$:* (i) *$T_D^j$ maps $C(I^j)$ into itself;* (ii) *T_D^j is a contraction on $(C(I^j), \|\cdot\|_\infty)$ with modulus $\beta(D)$;* (iii) *there exists a unique $V_D^j \in C(I^j)$ such that $V_D^j = T_D^j V_D^j$; and* (iv) *the induced policy correspondence in K' is nonempty, compact-valued, and upper hemi-continuous.*

The branch-restricted values V_D^{low} and V_D^{high} are the section's main local dynamic objects. They are constrained continuation values on admissible neighborhoods, not global value functions for the unrestricted planner problem. To assess how informative the local branch analysis is for the unrestricted problem near the switching region, we also solve a reduced Bellman problem on a wide interval $I_D^{\text{glob}} \supset I^{\text{low}} \cup I^{\text{high}}$ and compare the resulting unrestricted value V_D^{glob} with the branch envelope

$$V_D^{\text{env}}(K) \equiv \max\{V_D^{\text{low}}(K), V_D^{\text{high}}(K)\}, \qquad K \in J. \tag{6.7}$$

That cross-check remains numerical validation, not theory.

6.3 Activation sets, activation pockets, and first crossings

The key overlap object is the branch value gap

$$\Delta(D, K) \equiv V_D^{\text{high}}(K) - V_D^{\text{low}}(K), \qquad K \in J. \tag{6.8}$$

In the present environment the right dynamic objects are not, *a priori*, globally unique Skiba points. They are the subsets of the overlap where the high continuation weakly dominates.

Definition 6.5 (Activation objects)**.** *For fixed D, define the threshold set*

$$T(D) \equiv \{K \in J : \Delta(D, K) = 0\}, \tag{6.9}$$

the activation set

$$A(D) \equiv \{K \in J : \Delta(D, K) \geq 0\}, \tag{6.10}$$

and, when $A(D) \neq \varnothing$, the minimal activation threshold

$$K_S^{\min}(D) \equiv \inf A(D). \tag{6.11}$$

If $A(D) \neq \varnothing$, the connected components of $A(D)$ are the activation pockets. Ordered by left endpoints, write them as $P_1(D), P_2(D), \ldots$ Let $P_1(D) = [\underline{P}^1(D), \overline{P}^1(D)]$ denote the first pocket and define its width by

$$W_1(D) \equiv \overline{P}^1(D) - \underline{P}^1(D). \tag{6.12}$$

If $K_S^{\min}(D) = \underline{J}$, where $\underline{J} \equiv \inf J$, the threshold is boundary-attached. In that case, provided $T(D) \cap (\underline{J}, \overline{J}] \neq \varnothing$, define the first interior crossing by

$$K^{(1)}(D) \equiv \inf\{K > \underline{J} : \Delta(D, K) = 0\}. \tag{6.13}$$

Proposition 6.6 (Activation geometry and pocket decomposition)**.** *Fix $D \in \mathcal{D}$.* (i) *$\Delta(D, \cdot)$ is continuous on J. Hence $T(D)$ and $A(D)$ are closed.* (ii) *If $A(D) \neq \varnothing$, then $K_S^{\min}(D) \in A(D)$ is well defined.* (iii) *Every connected component of $A(D)$ is a compact interval. In particular, the activation pockets are compact intervals.* (iv) *If $\Delta(D, \underline{J}) < 0 < \Delta(D, \overline{J})$, then $T(D) \neq \varnothing$.* (v) *If $\Delta(D, \cdot)$ is strictly monotone on J, then $T(D)$ is a singleton and the model has a unique local indifference point on the overlap.*

The activation set is the primitive object. When $A(D)$ is boundary-attached, $K_S^{\min}(D) = \underline{J}$ is partly pinned by the chosen lower edge of the high branch and should not be treated as an intrinsic economic threshold. In that case the economically informative quantities are the *leftmost interior crossing* $K^{(1)}(D)$, the upper endpoint of the first activation pocket, and the associated width $W_1(D)$. This distinction is essential in the benchmark below, where the value gap is multi-crossing under nearby branch constructions.

6.4 Local comparative statics in the wedge

The critical-capital literature proves global continuity and monotonicity results in more restrictive environments. Here the right analogue is local and branch-disciplined.

Proposition 6.7 (Local Lipschitz continuity of branch values)**.** *Fix a compact wedge set $\mathcal{D} \subset (0, \infty)$ and an admissible local branch pair $(I^{\text{low}}, I^{\text{high}})$ on $\mathcal{D}$. Let*

$$\bar{\beta} \equiv \sup_{D \in \mathcal{D}} \beta(D) < 1, \qquad M_R^j \equiv \sup_{(K,K') \in I^j \times I^j} |R(K, K')| < \infty.$$

Then for each branch $j \in \{\text{low}, \text{high}\}$ and any $D, \widetilde{D} \in \mathcal{D}$,

$$\|V_D^j - V_{\widetilde{D}}^j\|_\infty \leq \frac{M_R^j}{(1-\bar{\beta})^2} |\beta(D) - \beta(\widetilde{D})|. \tag{6.14}$$

Hence $(D, K) \mapsto V_D^j(K)$ is jointly continuous on $\mathcal{D} \times I^j$.

Corollary 6.8 (Joint continuity of the branch gap)**.** *Under the assumptions of Proposition 6.7, the gap $(D, K) \mapsto \Delta(D, K)$ is jointly continuous on $\mathcal{D} \times J$. In particular, the activation correspondence $D \mapsto A(D)$ has closed graph.*

The explicit bound in (6.14) matters. It says more than "standard fixed-point continuity": once the branch neighborhoods are fixed, the branch values move at most linearly with the wedge through the discount factor. That is the appropriate local analogue of the continuity results in Akao et al. [2011, 2025]. It also avoids pretending that a global critical-capital stock exists for every D.

The next proposition isolates the exact condition under which a first-pocket endpoint can be followed continuously. The statement is monotonicity-based rather than derivative-based. This is deliberate. In dynamic programming, differentiability of the value gap is a theorem rather than a notation choice [Benveniste and Scheinkman, 1979], and the present paper does not need to assume more than local single crossing.

Proposition 6.9 (Local persistence of the first crossing)**.** *Assume the setup of Corollary 6.8. Fix $D_0 \in \mathcal{D}$. Suppose there exist $a < b$ in J and a neighborhood $\mathcal{N} \subset \mathcal{D}$ of D_0 such that* (i) *for every $D \in \mathcal{N}$, the map $K \mapsto \Delta(D, K)$ is strictly decreasing on $[a, b]$; and* (ii) *$\Delta(D_0, a) > 0 > \Delta(D_0, b)$. Then there exists a unique continuous function $K^\times : \mathcal{N} \to (a, b)$ such that*

$$\Delta(D, K^\times(D)) = 0 \qquad \textit{for all } D \in \mathcal{N}. \tag{6.15}$$

If, in addition, the threshold is boundary-attached on $\mathcal{N}$ and $[a, b]$ contains the first sign reversal after $\underline{J}$, then $K^\times(D) = K^{(1)}(D) = \overline{P}^1(D)$ on $\mathcal{N}$.

Proposition 6.9 is the paper's main local dynamic selection theorem. It is intentionally conditional. The benchmark below should not be read as a global single-crossing calibration; the overlap gap exhibits additional sign reversals beyond the leftmost crossing under nearby branch constructions. The economically robust benchmark object is therefore the leftmost interior crossing and the width of the first activation pocket, not a claim that the entire overlap is governed by a single global threshold.

Supplement to Proposition 6.9 (stable leftmost crossing). Suppose, in addition, that there exist $a_0 < a < b$ and a neighborhood $\mathcal{N}_0 \subset \mathcal{N}$ of D_0 such that $\Delta(D, K) > 0$ for every $D \in \mathcal{N}_0$ and every $K \in [\underline{J}, a_0]$. Then the continuous zero selected by Proposition 6.9 is the *leftmost* interior crossing on $\mathcal{N}_0$: for every $D \in \mathcal{N}_0$,

$$K^\times(D) = K^{(1)}(D).$$

Thus a local single-crossing window together with a uniform positive left buffer is enough to make the leftmost-crossing object intrinsic on that neighborhood even when the global overlap remains multi-crossing.

Proof. For any $D \in \mathcal{N}_0$, the positivity condition on $[\underline{J}, a_0]$ rules out zeros of $\Delta(D, \cdot)$ to the left of a_0. Proposition 6.9 yields a unique zero $K^{\times}(D) \in (a, b)$. Because no earlier zero exists, this zero is the first zero after $\underline{J}$, which is exactly the definition of $K^{(1)}(D)$. □

6.5 Benchmark: large stationary separation, boundary-attached entry diagnostic, and a narrow leftmost crossing

Under the baseline branch construction, the high-branch interval begins at

$$\underline{I}^{\text{high}} = 1.3600 \times 10^{-3},$$

while the low steady state is approximately

$$K_L \approx 1.35 \times 10^{-3}.$$

Dense overlap evaluation shows that the value gap is already nonnegative at the lower edge of the high branch under both low-branch truncations. Accordingly,

$$K_S^{\min} = \underline{I}^{\text{high}} = 1.3600 \times 10^{-3}.$$

Because this object is boundary-attached, however, it is reported only as a *boundary diagnostic*. The economic interpretation falls instead on the leftmost interior crossing

$$K^{(1)}(D) \in [1.36139,\, 1.36145] \times 10^{-3},$$

which lies only slightly above the lower edge of the high branch, and on the associated first-pocket width

$$W_1(D) \in [1.39,\, 1.45] \times 10^{-6}.$$

By contrast, the full stationary capital gap remains huge:

$$\frac{K_H - K_L}{C_L} \approx 171.4.$$

The benchmark message is therefore not a large one-time big push. It is *large stationary separation together with a narrow leftmost crossing region.* Measured against the low steady state, the first-pocket width is only about 0.10% of K_L, whereas $K_H/K_L \approx 6.19$ implies a stationary capital gap above 500% of K_L.

This left-edge interpretation is also the one most stable to nearby branch constructions. Perturbing the lower bound of the high branch moves the boundary-attached diagnostic $K_S^{\min}$ nearly one-for-one, whereas the leftmost interior crossing remains extremely close to that boundary. At the same time, the overlap gap exhibits multiple sign changes. The benchmark should therefore be described as a multi-crossing local-activation environment whose stable quantitative message is the narrowness of the first crossing region, not the level of the boundary-attached minimum.

Appendix B reports the full nearby-endpoint robustness grid. When the lower edge of the high branch is moved from 1.355×10^{-3} to 1.365×10^{-3}, the boundary-attached diagnostic $K_S^{\min}$ moves nearly one-for-one with that endpoint, while the leftmost interior crossing remains extremely close

Low interval	$\Delta(\underline{I}^{\text{high}})$	$K^{(1)}$	W_1	$K_S^{\min}$ (diagnostic)	Sign changes	$\|TV^j - V^j\|_\infty$
$[0.0009, 0.0026]$	9.31×10^{-6}	1.36145×10^{-3}	1.45×10^{-6}	1.3600×10^{-3}	6	9.53×10^{-9}
$[0.0009, 0.0030]$	8.84×10^{-6}	1.36139×10^{-3}	1.39×10^{-6}	1.3600×10^{-3}	6	9.27×10^{-9}

Table 4: Benchmark local dynamic diagnostics. The boundary-attached object $K_S^{\min}$ is reported only as a branch-boundary diagnostic; the economic emphasis falls on the leftmost interior crossing $K^{(1)}$ and the first-pocket width W_1.

to the new boundary and the overlap gap continues to display multiple sign changes. In the reported runs the number of sign changes ranges from 6 to 12, which is why the benchmark should be described as a multi-crossing activation-correspondence environment rather than as a clean single-crossing calibration.

6.6 Unrestricted reduced-Bellman validation

Branch-restricted values are constrained continuation values. The unrestricted solve is therefore asked a narrower question: does the branch envelope approximate the unrestricted planner problem *near the leftmost crossing*? We organize the cross-check around four objects: the unrestricted leftmost interior crossing $K_{\text{glob}}^{(1)}$, the envelope gap

$$E(K) \equiv V^{\text{glob}}(K) - V^{\text{env}}(K), \qquad K \in J,$$

the policy discrepancies $D_{K'}$ and D_L, and two stronger decision-quality objects defined in Appendix B, the envelope-policy value loss $L^{\text{env}\to\text{glob}}$ and the off-grid Bellman optimality gap $O(K)$. The last two are the right diagnostics in a threshold problem because small value errors can still misplace the switching boundary.

Under the preferred wide-low specification, the unrestricted solve leaves the headline message intact. The leftmost interior crossing remains extremely close to the lower edge of the high branch:

$$K_{\text{glob}}^{(1)} \approx 1.36157 \times 10^{-3}.$$

The envelope is close to the unrestricted planner value but does not dominate it pointwise:

$$\|E\|_{\infty,J} \approx 8.97 \times 10^{-3}, \qquad \min_{K\in J} E(K) \approx -2.90 \times 10^{-4}.$$

The policy comparison is more favorable in capital than in labor:

$$D_{K'} \approx 1.46 \times 10^{-5}, \qquad D_L \approx 1.75 \times 10^{-2}.$$

These raw diagnostics are not all equally load-bearing for the paper's actual claim. The unrestricted solve is not being asked to deliver pointwise global policy identity. It is being asked to validate three narrower objects: preservation of a left-edge crossing, approximate branch ranking near that crossing, and absence of a material welfare loss from using the branch envelope in that region.

On those margins the unrestricted run is materially stronger than the raw table alone suggests. Relative to the two preferred branch benchmarks $K^{(1)} \in [1.36139, 1.36145] \times 10^{-3}$, the unrestricted

crossing differs by only $(1.2\text{–}1.8) \times 10^{-7}$, i.e. roughly 0.009–0.013% of K_L. Measured against the first-pocket width itself, the unrestricted crossing lies only about 8–13% of one pocket width above the local first-pocket endpoint. Thus the unrestricted and branch-restricted calculations agree on the leftmost crossing at a substantially finer scale than the width of the switching region they are meant to identify.

With log utility, the envelope gap can also be translated into permanent consumption-equivalent units at fixed labor:

$$\lambda_E(K) \equiv \exp\left(\frac{(1-\beta)E(K)}{\eta}\right) - 1.$$

Under the benchmark calibration, $\|E\|_{\infty,J}$ corresponds to about 8.97×10^{-4} (0.0897%) in permanent consumption-equivalent terms, while the negative sliver $\min_{K \in J} E(K)$ corresponds to about -2.90×10^{-5} (-0.0029%). The capital-policy discrepancy is also small relative to the full stationary capital gap:

$$\frac{D_{K'}}{K_H - K_L} \approx 2.1 \times 10^{-3}.$$

By contrast, the labor discrepancy is the least favorable unrestricted metric; relative to the benchmark steady-state labor levels it is about 5–7%. That labor margin is exactly why the unrestricted solve is used here to validate the left-edge crossing geometry and approximate value ranking, not to claim pointwise global policy equivalence.

The right interpretation is therefore more specific than "encouraging but incomplete." Against the paper's claim-matching validation targets, the unrestricted run is already supportive: the leftmost crossing is preserved to within a small fraction of one pocket width, the envelope gap is economically tiny in permanent consumption-equivalent units, and the capital policy remains extremely close in the only state dimension that governs continuation. What remains incomplete is unrestricted pointwise policy equivalence, especially on the labor margin. The thin negative envelope sliver and the larger labor discrepancy therefore limit how far the dynamic claims can be pushed, but they no longer undermine the paper's actual local-activation message. Appendix B summarizes the current implementation choices, the nearby-endpoint robustness exercise, and the reporting standards that should guide the next unrestricted rerun.

Comparison	$K^{(1)}_{\text{glob}}$	$\|E\|_{\infty,J}$	$\min_{K \in J} E(K)$	$D_{K'}$	D_L
Preferred unrestricted run	1.36157×10^{-3}	8.97×10^{-3}	-2.90×10^{-4}	1.46×10^{-5}	1.75×10^{-2}

Validation diagnostics. The unrestricted run is interpreted as validation rather than theorem. Appendix B defines the stronger decision-quality objects $L^{\text{env}\to\text{glob}}$ and $O(K)$ for the next adaptive refinement pass.

6.7 Two wedge thresholds

The wedge thresholds are steady-state existence objects and should be distinguished sharply from the local dynamic selection objects above.

Definition 6.10 (Fold threshold)**.** *Let r^* denote the maximizer of $\phi_K(r)$. The* fold threshold D_F *is defined by*

$$G(r^*; D_F) = 0.$$

At D_F, the low-r and high-r steady states collide at the fold.

Definition 6.11 (Final saddle-node threshold)**.** *If same-branch multiplicity survives beyond the fold, define D_{SN} and $r_{SN} > r^*$ by*

$$G(r_{SN}; D_{SN}) = 0, \qquad G_r(r_{SN}; D_{SN}) = 0.$$

At D_{SN}, the remaining right-branch pair annihilates.

For the wedge taxonomy, impose the following regularity conditions on the relevant feasible set of (r, D) pairs: $G(r; D)$ is continuous in (r, D), continuously differentiable in r, interior roots are simple except at regular tangencies, and for $D < D_F$ sufficiently close to D_F the economy has one simple root on each branch. If a same-branch pair survives beyond the fold, assume that it disappears through a regular right-branch saddle-node satisfying the conditions above.

Proposition 6.12 (Two-threshold bifurcation structure)**.** *Under the regularity conditions just stated, $D_F \leq D_{SN}$ and the wedge taxonomy is*

$$\begin{aligned} D < D_F &\quad : \textit{opposite-branch multiplicity,} \\ D_F < D < D_{SN} &\quad : \textit{possible same-branch multiplicity,} \\ D > D_{SN} &\quad : \textit{no interior steady state.} \end{aligned}$$

If no same-branch pair survives beyond the fold, then $D_F = D_{SN}$.

Proof. At $D = D_F$, opposite-branch roots meet at (r^*, D_F) by construction. If a same-branch pair remains for larger D, the regularity assumptions imply that its disappearance must occur at a tangency satisfying the saddle-node conditions above. If no such pair survives, the disappearance of opposite-branch multiplicity is already the final disappearance, so $D_F = D_{SN}$. □

In the benchmark calibration,

$$\frac{D_F}{D_0} \approx 1.023, \qquad \frac{D_{SN}}{D_0} \approx 1.031.$$

Thus a wedge increase of about 2.3% eliminates opposite-branch multiplicity, while an increase of about 3.1% eliminates the last interior steady states. Under the financing map $D(s) = D_0 + s/\beta$, these benchmark ratios correspond to spread increases of roughly 27 and 36 basis points, respectively. The disappearance window is therefore economically visible even though it remains narrow in proportional wedge terms.

6.8 Local recovery map

The wedge thresholds classify existence; the activation objects classify local selection. The natural recovery set is therefore

$$R^{\text{loc}} \equiv \{(D, K) \in \mathcal{D} \times J : D < D_{SN},\ K \in A(D)\}.$$

This is a *local branch-comparison object on the overlap J*, not a globally verified basin map for the unrestricted planner problem.

Proposition 6.13 (Two-dimensional local recovery map)**.** *Fix a compact wedge set $\mathcal{D}$ on which the local branch pair is admissible and define R^{loc} as above. Then:* (i) *if $D \geq D_{SN}$, the high local continuation is unavailable because the upper interior branch no longer exists;* (ii) *if $D < D_{SN}$ and $K \in A(D)$, the local branch comparison weakly favors the high continuation at state K; and* (iii) *if $D_F < D < D_{SN}$, steady-state coexistence and local dynamic activation need not coincide: the high continuation may exist locally while failing to be selected from some initial capital stocks in J. If the relevant activation set is boundary-attached, the operational recovery condition is better summarized by the leftmost interior crossing or first-pocket endpoint than by $K_S^{\min}$ itself.*

Recovery is therefore genuinely two-dimensional. Lowering the wedge enough to restore the existence of the upper branch is not sufficient if the capital stock remains outside the relevant local activation region; conversely, raising capital alone is not enough if the wedge remains above the disappearance threshold. In a fully refined quantitative implementation, the right object to plot is a shaded local recovery map in (D, K) space rather than a single schematic threshold curve.

7 Conclusion

This paper's main global contribution is steady-state. The scalar diagnostic $G(r; D)$ turns the interior steady-state problem into one-dimensional root finding and yields a closed-form minimum viable scale. In that respect the paper already delivers a durable result: the fold regime makes increasing-returns Ramsey dynamics analytically tractable without restoring global concavity. The same reduction also sharpens the auxiliary feasibility condition behind multiplicity: along the Euler locus, consumption positivity can be written directly in primitive form, and in the benchmark it is automatic because $D_0 > \delta\theta$.

The dynamic contribution is intentionally local. On compact admissible branch neighborhoods the reduced Bellman problem is well posed, activation sets admit a pocket decomposition, and under local single-crossing conditions the leftmost interior crossing varies continuously with the wedge. In the benchmark, however, the minimal local activation threshold is boundary-attached at

$$K_S^{\min} = 1.3600 \times 10^{-3},$$

so it is best read as a branch-boundary diagnostic rather than as an intrinsic economic threshold. The economically informative local object is the first interior crossing,

$$K^{(1)} \in [1.36139,\, 1.36145] \times 10^{-3},$$

which lies only slightly above that boundary. Measured against the low steady state, the benchmark boundary-attached threshold is only about 0.95% higher than K_L and roughly 31.5% of one period of low-state consumption. The quantitative message is therefore not a giant one-shot big push. It is large stationary separation together with a very narrow first activation pocket.

The wedge side is equally sharp. In the benchmark calibration,

$$\frac{D_F}{D_0} \approx 1.023, \qquad \frac{D_{SN}}{D_0} \approx 1.031.$$

A wedge increase of roughly 2.3% eliminates opposite-branch multiplicity, while an increase of roughly 3.1% eliminates the last interior steady states. Under the financing map $D(s) = D_0 + s/\beta$, these benchmark movements correspond to spread increases of roughly 27 and 36 basis points. Recovery is therefore genuinely two-dimensional: lowering the wedge enough to restore the upper branch is not sufficient if the economy's capital stock remains outside the local activation region, and a capital injection alone is not enough if the wedge remains above the disappearance threshold. The operative recovery object is the local set

$$R^{\text{loc}} = \{(D, K) : D < D_{SN},\ K \in A(D)\},$$

or, near the lower edge when the first pocket is boundary-attached, the corresponding first-pocket endpoint $K^{(1)}$.

The unrestricted reduced-Bellman exercise remains a validation step, not a theorem. In the current stable first-pass run,

$$\|E\|_{\infty,J} \approx 8.97 \times 10^{-3}, \qquad \min_{K \in J} E(K) \approx -2.90 \times 10^{-4},$$

$$D_{K'} \approx 1.46 \times 10^{-5}, \qquad D_L \approx 1.75 \times 10^{-2}.$$

Taken against the paper's actual validation targets, those numbers are already supportive rather than merely suggestive. The unrestricted crossing differs from the preferred local crossing by only about 0.009–0.013% of K_L, or roughly 8–13% of one first-pocket width. In permanent consumption-equivalent terms, $\|E\|_{\infty,J}$ is below one-tenth of one percent and the negative envelope sliver is about three-thousandths of one percent in absolute value. The capital-policy discrepancy is only about 0.21% of the full stationary capital gap. The remaining qualification comes from the labor margin: D_L is materially larger, so the unrestricted solve still falls short of pointwise global policy equivalence. Accordingly, the unrestricted run now does enough to validate the local activation story the paper actually tells, while still not upgrading it into a global selection theorem.

The benchmark is illustrative rather than structurally estimated. Its role is to show that the model's core objects — $L_{\min}$, the scalar diagnostic, the activation correspondence, the leftmost interior crossing, and the wedge thresholds — are quantitatively meaningful in a plausible neighborhood. Appendix B records the current implementation choices and the nearby-endpoint robustness results, and it makes explicit what the next rerun should deliver: adaptive refinement around every sign change of the branch-value gap, interpolation-robust unrestricted validation on a common global grid, off-grid Bellman optimality gaps and envelope-policy losses near the switching region, and computed recovery maps and transition paths rather than schematic threshold drawings.

The broader lesson is that increasing returns need not force a choice between elegance and economic relevance. In the present Ramsey–CES environment, the steady-state side admits a sharp global scalar theory, while the dynamic side admits a disciplined local theory of activation and hysteresis. That combination is enough to characterize when a high-scale regime exists, when it is locally selected, how financing conditions reshape the recovery region, and why restoring a destroyed high regime can require both a wedge reversal and a capital intervention.

Appendix A. Proofs for Section 6

A.1 Proof of Proposition 6.2

Fix compact intervals $I, I' \subset (0, B/\delta]$ and suppose (I, I') is uniformly strictly feasible. By Definition 6.1, there exists $\bar{L} \in (0,1)$ such that

$$c(I, I'; \bar{L}) = \inf_{K \in I,\, K' \in I'} C(K, \bar{L}, K') > 0.$$

Hence the benchmark payoff

$$u \equiv \inf_{K \in I,\, K' \in I'} \Big[\eta \log C(K, \bar{L}, K') + (1-\eta)\log(1-\bar{L})\Big]$$

is finite. Also, continuity of C on the compact set $I \times [0,1] \times I'$ implies

$$\bar{C} \equiv \sup_{K \in I,\, L \in [0,1],\, K' \in I'} C(K, L, K') < \infty.$$

Choose $\varepsilon \in (0, 1-\bar{L})$ small enough that

$$\eta \log \varepsilon < u - 1 \qquad \text{and} \qquad \eta \log \bar{C} + (1-\eta)\log \varepsilon < u - 1. \tag{A.1}$$

Consider any feasible triple (K, L, K') with $(K, K') \in I \times I'$. If $C(K, L, K') < \varepsilon$, then

$$\eta \log C(K, L, K') + (1-\eta)\log(1-L) \le \eta \log \varepsilon < u.$$

If instead $1 - L < \varepsilon$, then

$$\eta \log C(K, L, K') + (1-\eta)\log(1-L) \le \eta \log \bar{C} + (1-\eta)\log \varepsilon < u.$$

Either case yields a payoff strictly below the benchmark payoff attainable at $(K, \bar{L}, K')$. Therefore no maximizer can satisfy $C(K, L, K') < \varepsilon$ or $1 - L < \varepsilon$. Every maximizer must lie in $\Gamma_\varepsilon(K, K')$ defined in (6.5).

For each $(K, K') \in I \times I'$, the set $\Gamma_\varepsilon(K, K')$ is nonempty because it contains $\bar{L}$, compact because it is a closed subset of $[0, 1-\varepsilon]$, and varies upper hemi-continuously with (K, K') because it is defined by a continuous inequality. The period payoff is continuous on the graph of Γ_ε. Berge's maximum theorem therefore implies that $R(K, K')$ is finite and continuous on $I \times I'$, and that the argmax correspondence is nonempty, compact-valued, and upper hemi-continuous. Part (iii) follows immediately because $(K, K') \mapsto \beta(D)W(K')$ is continuous whenever W is continuous.

A.2 Proof of Proposition 6.4

Fix $D \in \mathcal{D}$ and $j \in \{\text{low}, \text{high}\}$. Since I^j is admissible, Proposition 6.2 implies that $(K, K') \mapsto R(K, K')$ is continuous on $I^j \times I^j$. Therefore, for any $W \in C(I^j)$, the objective

$$Q_D^j(K, K'; W) \equiv R(K, K') + \beta(D)W(K')$$

is continuous on the compact set $I^j \times I^j$. Maximization over the compact control set I^j then shows that $T_D^j W \in C(I^j)$, proving part (i), and that the policy correspondence is nonempty, compact-valued, and upper hemi-continuous by Berge's theorem, proving part (iv).

For any $W_1, W_2 \in C(I^j)$ and any $K \in I^j$,

$$|(T_D^j W_1)(K) - (T_D^j W_2)(K)| \leq \beta(D)\|W_1 - W_2\|_\infty.$$

Taking suprema over K gives

$$\|T_D^j W_1 - T_D^j W_2\|_\infty \leq \beta(D)\|W_1 - W_2\|_\infty.$$

Since $\beta(D) \in (0, 1)$, T_D^j is a contraction. Banach's fixed-point theorem yields a unique fixed point $V_D^j \in C(I^j)$, proving part (iii).

A.3 Proof of Proposition 6.6

By Proposition 6.4, each branch value V_D^j is continuous on I^j, hence the difference $\Delta(D, \cdot)$ is continuous on the compact interval J. This proves part (i). If $A(D) \neq \varnothing$, then $A(D)$ is a nonempty closed subset of a compact set, hence compact. Therefore $\inf A(D) \in A(D)$, proving part (ii).

For part (iii), every connected subset of the real line is an interval. Since $A(D)$ is a closed subset of the compact interval J, each connected component of $A(D)$ is a closed and bounded interval, hence compact. Part (iv) is the intermediate value theorem applied to the continuous map $K \mapsto \Delta(D, K)$. Part (v) follows because a strictly monotone continuous function on an interval has at most one zero.

A.4 Proof of Proposition 6.7 and Corollary 6.8

Fix a branch j. Let $M_R^j = \sup_{I^j \times I^j} |R| < \infty$ and $\bar{\beta} = \sup_{D \in \mathcal{D}} \beta(D) < 1$. Since $V_{\widetilde{D}}^j = T_{\widetilde{D}}^j V_{\widetilde{D}}^j$,

$$\|V_{\widetilde{D}}^j\|_\infty = \|T_{\widetilde{D}}^j V_{\widetilde{D}}^j\|_\infty \leq M_R^j + \beta(\widetilde{D})\|V_{\widetilde{D}}^j\|_\infty \leq M_R^j + \bar{\beta}\|V_{\widetilde{D}}^j\|_\infty,$$

so

$$\|V_{\widetilde{D}}^j\|_\infty \leq \frac{M_R^j}{1 - \bar{\beta}}. \tag{A.2}$$

Now compare V_D^j and $V_{\widetilde{D}}^j$:

$$\begin{aligned}\|V_D^j - V_{\widetilde{D}}^j\|_\infty &= \|T_D^j V_D^j - T_{\widetilde{D}}^j V_{\widetilde{D}}^j\|_\infty \\ &\le \|T_D^j V_D^j - T_D^j V_{\widetilde{D}}^j\|_\infty + \|T_D^j V_{\widetilde{D}}^j - T_{\widetilde{D}}^j V_{\widetilde{D}}^j\|_\infty \\ &\le \beta(D)\|V_D^j - V_{\widetilde{D}}^j\|_\infty + |\beta(D) - \beta(\widetilde{D})|\,\|V_{\widetilde{D}}^j\|_\infty.\end{aligned}$$

Using (A.2) and $1 - \beta(D) \ge 1 - \bar{\beta}$ gives

$$\|V_D^j - V_{\widetilde{D}}^j\|_\infty \le \frac{M_R^j}{(1-\bar{\beta})^2}\,|\beta(D) - \beta(\widetilde{D})|.$$

This proves (6.14). Joint continuity of $(D, K) \mapsto V_D^j(K)$ follows immediately because the bound is uniform in K. The corollary follows by taking the difference between the two jointly continuous branch values. For the closed-graph statement, let $(D_n, K_n) \to (D, K)$ with $K_n \in A(D_n)$. Then $\Delta(D_n, K_n) \ge 0$ for all n, and continuity implies $\Delta(D, K) \ge 0$, so $K \in A(D)$.

A.5 Proof of Proposition 6.9

Fix $D \in \mathcal{N}$. By continuity of $K \mapsto \Delta(D, K)$ and the sign conditions at a and b, the intermediate value theorem implies the existence of at least one $K^\times(D) \in (a, b)$ with $\Delta(D, K^\times(D)) = 0$. Strict decrease of $\Delta(D, \cdot)$ on $[a, b]$ implies that this zero is unique.

To prove continuity, let $D_n \to D$ in $\mathcal{N}$ and write $K_n \equiv K^\times(D_n)$. Since $[a, b]$ is compact, every subsequence of $\{K_n\}$ has a further subsequence converging to some $\bar{K} \in [a, b]$. By joint continuity of Δ,

$$0 = \lim_{m\to\infty} \Delta(D_{n_m}, K_{n_m}) = \Delta(D, \bar{K}).$$

Uniqueness of the zero on $[a, b]$ implies $\bar{K} = K^\times(D)$. Hence every convergent subsequence of $\{K_n\}$ has the same limit, so $K_n \to K^\times(D)$. Therefore $K^\times$ is continuous on $\mathcal{N}$.

If the threshold is boundary-attached on $\mathcal{N}$ and $[a, b]$ contains the first sign reversal after $\underline{J}$, then the unique zero in $[a, b]$ is, by construction, both the upper endpoint of the first activation pocket and the first interior crossing. Thus $K^\times(D) = K^{(1)}(D) = \overline{P}^1(D)$ on $\mathcal{N}$.

A.6 Proof of Proposition 6.13

Part (i) follows from the definition of D_{SN} as the final wedge at which interior steady states disappear. For any $D \ge D_{SN}$, the high interior branch is absent, so no local high continuation exists to be selected. Part (ii) is immediate from the definition of the activation set: $K \in A(D)$ means exactly that the high branch value weakly exceeds the low branch value at state K. Part (iii) follows because D_F and D_{SN} are distinct objects. Between them, the steady-state side may still support a surviving interior branch while the local value comparison need not favor that branch for every capital stock in the overlap. Therefore the recovery condition is joint in (D, K) rather than

reducible to either coordinate alone.

Appendix B. Numerical verification, unrestricted validation, and reporting standards

This appendix records the benchmark implementation that underlies the revised local-dynamic claims in Section 6 and makes the validation philosophy explicit. The main numerical objects are the branch-restricted continuation values, the dense-overlap value gap, the first activation pocket, and the unrestricted reduced-Bellman cross-check. In a nonconcave threshold problem the main accuracy objects are Bellman and policy losses, not Euler equations alone [Judd, 1992, Santos and Vigo-Aguiar, 1998]. We therefore organize the solver checks in layers: branch Bellman residuals, unrestricted envelope gaps, local policy discrepancies, and off-grid Bellman checks on refined control meshes near the switching region. For the next unrestricted rerun, an upper-envelope-aware implementation in the spirit of Fella [2014] and Druedahl and Jorgensen [2017] is the natural benchmark upgrade.

B.1 Baseline branch-restricted solver

The baseline branch construction uses the common high-branch interval

$$I^{\text{high}} = [0.00136, 0.0120],$$

together with two low-branch truncations,

$$I^{(1)}_{\text{low}} = [0.0009, 0.0026], \qquad I^{(2)}_{\text{low}} = [0.0009, 0.0030].$$

For the low branch, state and control grids are nonuniform and concentrated near K_L; for the high branch, nodes are concentrated near the lower edge and around K_H. The baseline grids use $(n_K, n_{K'}) = (121, 361)$ on each low interval and $(141, 401)$ on the high interval. For every (K, K') pair on a branch grid, the static labor problem is solved directly, and the Bellman step maximizes over the discrete control grid. Baseline continuation values are interpolated with shape-preserving cubic Hermite interpolation (PCHIP), and the overlap value gap is evaluated on a dense 6,000-point grid. As a cross-check, the same exercise is repeated with linear interpolation.

Low interval	n^L_K	$n^L_{K'}$	n^H_K	$n^H_{K'}$	$\|TV^{\text{low}} - V^{\text{low}}\|_\infty$	$\|TV^{\text{high}} - V^{\text{high}}\|_\infty$	$\Delta(\underline{J})$	$K^{(1)}$
$[0.0009, 0.0026]$	121	361	141	401	9.53×10^{-9}	9.22×10^{-9}	9.31×10^{-6}	1.36145×10^{-3}
$[0.0009, 0.0030]$	121	361	141	401	9.27×10^{-9}	9.22×10^{-9}	8.84×10^{-6}	1.36139×10^{-3}

Table 5: Baseline branch-restricted solver diagnostics. Bellman residuals are tiny in every baseline specification, the value gap is positive at the lower edge of the overlap under both low-branch truncations, and the leftmost interior crossing remains extremely close to that lower edge.

B.2 High-branch lower-bound robustness

Because the benchmark threshold is boundary-attached, perturbing the lower bound of the high-branch interval is part of the object being reported rather than a cosmetic exercise. Table B.2 therefore varies the high-branch lower bound over a narrow neighborhood while holding the rest of the branch construction fixed.

Low interval	$\underline{I}^{\text{high}}$	$\Delta(\underline{J})$	$K_S^{\min}$	$K^{(1)}$	Sign changes of Δ
$[0.0009, 0.0026]$	1.355×10^{-3}	2.04×10^{-6}	1.355×10^{-3}	1.35545×10^{-3}	8
$[0.0009, 0.0026]$	1.360×10^{-3}	9.31×10^{-6}	1.360×10^{-3}	1.36145×10^{-3}	6
$[0.0009, 0.0026]$	1.365×10^{-3}	1.81×10^{-5}	1.365×10^{-3}	1.36668×10^{-3}	12
$[0.0009, 0.0030]$	1.355×10^{-3}	2.55×10^{-6}	1.355×10^{-3}	1.35552×10^{-3}	8
$[0.0009, 0.0030]$	1.360×10^{-3}	8.84×10^{-6}	1.360×10^{-3}	1.36139×10^{-3}	6
$[0.0009, 0.0030]$	1.365×10^{-3}	1.69×10^{-5}	1.365×10^{-3}	1.36666×10^{-3}	12

Table 6: High-branch lower-bound robustness of local threshold diagnostics. The minimal threshold is boundary-attached throughout, so $K_S^{\min}$ moves nearly one-for-one with the lower bound of the high branch. At the same time, the leftmost interior crossing always remains very close to that boundary.

The message is disciplined rather than over-precise. The benchmark threshold is boundary-attached throughout, so $K_S^{\min}$ moves nearly one-for-one with the lower bound of the high branch. At the same time, the first interior crossing always remains very close to that boundary. In the linear-interpolation cross-check, $\Delta(\underline{J})$ remains positive and the first interior crossing moves only modestly upward, to roughly $[1.36180, 1.36340] \times 10^{-3}$ across the two low-branch truncations.

B.3 Unrestricted global cross-check

The unrestricted reduced Bellman problem is solved on the wide interval

$$I^{\text{glob}} = [0.0005, 0.05]$$

using a global union grid and linear interpolation. Table B.3 summarizes the currently stable first-pass diagnostics for the preferred wide-low specification.

Global interval	$K_{\text{glob}}^{(1)}$	$\|E\|_{\infty,J}$	$\min_{K \in J} E(K)$	$D_{K'}$	D_L
$[0.0005, 0.05]$	1.36157×10^{-3}	8.97×10^{-3}	-2.90×10^{-4}	1.46×10^{-5}	1.75×10^{-2}

Table 7: First-pass unrestricted reduced-Bellman cross-check. The restricted and unrestricted calculations tell a consistent qualitative story: the first interior crossing remains extremely close to the lower edge of the high branch, and the capital-policy disagreement is tiny.

The restricted and unrestricted calculations tell a consistent qualitative story: the first interior crossing remains extremely close to the lower edge of the high branch, and the capital-policy disagreement is tiny relative to the full stationary gap. The remaining negative envelope sliver and

the more noticeable labor-policy discrepancy justify the paper's restrained language on unrestricted policy equivalence, but they no longer overturn the local first-crossing geometry that the paper is actually trying to validate.

Three normalizations help interpret the remaining mismatch. First, relative to the two preferred local crossings $K^{(1)} \in [1.36139, 1.36145] \times 10^{-3}$, the unrestricted crossing differs by only $(1.2\text{–}1.8) \times 10^{-7}$, i.e. $0.9\text{–}1.3 \times 10^{-4}$ of K_L or about 8–13% of one first-pocket width. Second, with log utility the envelope gap can be translated into permanent consumption-equivalent units via

$$\lambda_E(E) \equiv \exp\left(\frac{(1-\beta)E}{\eta}\right) - 1.$$

Evaluated at $\|E\|_{\infty,J}$ and $\min_{K\in J} E(K)$, the corresponding magnitudes are 8.97×10^{-4} and -2.90×10^{-5}, respectively. Third, the capital-policy discrepancy satisfies $D_{K'}/(K_H - K_L) \approx 2.1 \times 10^{-3}$, whereas the labor-policy discrepancy remains on the order of 5–7% of benchmark steady-state labor.

Normalized diagnostic	Value
$\lvert K^{(1)}_{\text{glob}} - K^{(1)}\rvert / K_L$	$(0.9\text{–}1.3) \times 10^{-4}$
$\lvert K^{(1)}_{\text{glob}} - \overline{P}^{\,1}\rvert / W_1$	0.08–0.13
$D_{K'}/(K_H - K_L)$	2.1×10^{-3}
$\exp((1-\beta)\|E\|_{\infty,J}/\eta) - 1$	8.97×10^{-4}
$\exp((1-\beta)\min_{K\in J} E(K)/\eta) - 1$	-2.90×10^{-5}

Table 8: Normalized versions of the first-pass unrestricted diagnostics. The unrestricted crossing differs from the local first-pocket endpoint by only a small fraction of one pocket width, the capital-policy disagreement is tiny relative to the full stationary capital gap, and the envelope gap is economically tiny in permanent consumption-equivalent units even though the negative sliver keeps the interpretation local.

B.4 Additional validation objects

Two further diagnostics are useful for future refinements and for replication work. Let π^{env} denote the envelope policy on the overlap, and let π^{glob} denote the unrestricted global policy. The global value loss from using the envelope policy is

$$L^{\text{env}\to\text{glob}}(K) \equiv V^{\text{glob}}(K) - \Big[R\big(K, \pi^{\text{env}}(K)\big) + \beta(D)V^{\text{glob}}\big(\pi^{\text{env}}(K)\big)\Big]. \tag{B.1}$$

This quantity is economically more informative than a raw policy-distance metric because it measures the unrestricted value cost of following the branch-envelope decision rule.

The second object is an off-grid Bellman optimality gap evaluated on a refined control mesh:

$$O(K) \equiv \sup_{K'\in I^{\text{glob}}} \Big\{R(K,K') + \beta(D)V^{\text{glob}}(K')\Big\} - \Big\{R\big(K, \pi^{\text{glob}}(K)\big) + \beta(D)V^{\text{glob}}\big(\pi^{\text{glob}}(K)\big)\Big\}. \tag{B.2}$$

In a threshold problem these are the natural next-step diagnostics because they evaluate the quality

of the computed decisions directly. They are especially informative near the switching region, where interpolation error can matter even when Bellman residuals are already small.

B.5 Reporting standards for subsequent reruns

The revised dynamic section should be read together with a narrow reporting discipline. At a minimum, a subsequent unrestricted rerun should report

$$(K_L,\, K_S^{\min},\, K^{(1)},\, W_1,\, K_H,\, \#\mathrm{comp}(A),\, \|E\|_{\infty,J},\, D_F/D_0,\, D_{SN}/D_0),$$

together with the branch endpoints that generated those objects. That line converts the dynamic section from a suggestive discussion into a quantitative statement about local activation, pocket width, global-cross-check quality, and hysteresis.

The appendix table for the next rerun should therefore include, at minimum:

1. branch endpoints and overlap endpoints;
2. Bellman residuals for the low, high, and global solves;
3. maximum Euler residual away from a narrow neighborhood of the value crossing;
4. $K_S^{\min}$, $K^{(1)}$, and W_1;
5. the number of connected components of the activation set on the overlap;
6. $\|E\|_{\infty,J}$, $\min_{K\in J} E(K)$, and any normalized envelope-gap counterpart;
7. $L^{\mathrm{env}\to\mathrm{glob}}$ and $O(K)$ near the switching region; and
8. robustness to nearby branch endpoints, interpolation methods, and local mesh density.

The central numerical principle is simple: in a nonconcave problem with local kinks, threshold accuracy is the object. Bellman convergence by itself is not enough.

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